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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

AD2212-1 – TEXT AND SPEECH ANALYSIS
(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **QMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Text preprocessing in NLP typically includes:**
A) Tokenization, stemming, lemmatization, stop-word removal
B) Encryption, compression, hashing
C) Syntax tree parsing only
D) Image normalization
2. **Which of the following removes suffixes to reduce words to their root form, sometimes creating non-dictionary words?**
A) Lemmatization
B) Stemming
C) Tokenization
D) Stop-word removal
3. **Stop-words are typically removed in NLP because:**
A) They do not contribute significant meaning
B) They increase computational complexity without improving results
C) They are very frequent in the text
D) They provide little value for downstream learning tasks
4. **Bag of N-Grams model improves over BoW by:**
A) Considering word order in small windows
B) Removing stop-words automatically
C) Using embeddings
D) Always reducing dimensionality
5. **The main advantage of TF-IDF over BoW is:**
A) It considers importance of words relative to documents
B) It reduces dimensionality automatically
C) It ignores rare words completely
D) It requires deep learning models
6. **Word2Vec learns embeddings using:**
A) SVM classifiers
B) Frequency tables only
C) Decision trees
D) Neural networks with CBOW or Skip-gram architecture
7. **FastText improves over Word2Vec by:**
A) Using subword (character n-gram) information
B) Using larger corpora only
C) Removing stop-words automatically
D) Compressing embeddings
8. **Recurrent Neural Networks (RNNs) are mainly designed for:**
A) Fixed-length numerical input
B) Sequential data like text and speech
C) Image classification only
D) Static tabular data
9. **Knowledge-based QA systems typically use:**
A) Search engines only
B) Structured databases and ontologies
C) Word embeddings
D) Bag-of-words models only
10. **A classic QA model usually has which components?**
A) Query analysis, document retrieval, answer extraction
B) Speech synthesis, text translation, tokenization
C) Parsing only
D) Embedding generation only

11. **Dialogue system design typically includes:**
 - A) Natural Language Understanding (NLU), Dialogue Management, Natural Language Generation (NLG)
 - B) Tokenization, stemming, lemmatization
 - C) Only knowledge graphs
 - D) Rule-based syntax parsing only
12. **Which of the following is an example of a task-oriented dialogue system?**
 - A) Siri helping to set an alarm
 - B) Social chatbots like Replika
 - C) Wikipedia search
 - D) TF-IDF system
13. **End-to-end neural QA systems (like BERT-based models) differ from classic QA because:**
 - A) They rely only on IR
 - B) They jointly learn representation, retrieval, and answer extraction
 - C) They do not use embeddings
 - D) They avoid contextual understanding
14. **Text normalization in TTS involves:**
 - A) Converting numbers, abbreviations, and symbols into spoken form
 - B) Tokenizing text
 - C) Removing stop words
 - D) Lemmatization
15. **Prosody in speech synthesis refers to:**
 - A) Grammar correctness
 - B) Rhythm, stress, and intonation of speech
 - C) Tokenization of phonemes
 - D) Compression of speech signals
16. **Parametric TTS systems differ from concatenative systems because they:**
 - A) Use acoustic models to generate speech features instead of storing segments
 - B) Always sound perfectly natural
 - C) Do not use signal processing
 - D) Ignore prosody
17. **Acoustic modeling in speech recognition refers to:**
 - A) Modeling the relationship between phonemes and audio signals
 - B) Translating words into phonemes
 - C) Normalizing text input
 - D) Generating speech signals
18. **The purpose of feature extraction in speech recognition is to:**
 - A) Increase audio file size
 - B) Represent speech signals in a compact, informative form
 - C) Remove noise completely
 - D) Generate synthetic speech
19. **In HMMs, the "hidden" part refers to:**
 - A) Observed speech signals
 - B) Noise in the audio
 - C) Underlying sequence of phonemes or states not directly visible
 - D) Encrypted features
20. **The role of the language model in speech recognition is to:**
 - A) Map phonemes to waveforms
 - B) Generate synthetic voices
 - C) Normalize pitch and prosody
 - D) Predict the most likely word sequence given acoustic evidence

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the role of language, syntax, and structure in NLP with a suitable example.	8	L*2	1	1
	b) How to apply transformers to improve the handling of long-range dependencies in text and explain.	8	L3	2	1
2.	a) Explain the concept of tokenization, stop-words, stemming and lemmatization in NLP.	8	L2	1	1
	b) How to apply GloVe embeddings to analyze global co-occurrence statistics of words.	8	L3	2	1
3.	a) Explain the TF-IDF model.	8	L2	1	1
	b) How to apply FastText embeddings to handle words with prefixes and suffixes.	8	L3	2	1

Unit – II

4.	a)	Explain the role of knowledge-based question answering systems with a suitable example.	8	L2	3	1
	b)	Apply dialogue system design principles in creating a task-oriented assistant and explain.	8	L3	4	1
5.	a)	Explain the working of classic QA models.	8	L2	3	1
	b)	Apply TTS evaluation methods to compare quality across systems and explain with a suitable example.	8	L3	4	1
6.	a)	Explain the role of prosody in speech synthesis.	8	L2	3	1
	b)	Apply parametric synthesis methods to model human speech and explain.	8	L3	4	1

Unit – III

7.	a)	Explain the basic principles of Hidden Markov Models (HMM).	8	L2	5	1
	b)	Apply noise-robust methods to enhance speech recognition in real environments and explain.	8	L3	5	1
8.	a)	Explain the concept of language modelling in speech recognition with a suitable example.	8	L2	5	1
	b)	How to apply feature extraction methods to convert raw speech signals into useful inputs.	8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
